

Kaiyu Hou

Curriculum Vitae

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I am currently a Senior Engineer of Kangaroo (RunD) Network Group at Alibaba Cloud. My job focuses on improving networking foundations for cloud native systems and agentic sandboxes, aiming to provide flexible and efficient networks within containers and above cloud VPCs.

I received my Ph.D. degree in computer science in 2022 from Northwestern University, under the guidance of Prof. Yan Chen.

I have a broad interest in various aspects of cloud networks, network protocols, and networked systems. My dissertation work focuses on network optimization of microservices, serverless based clouds. I am also working for introducing formal methods into network protocol verification.

Education

- **Ph.D. in Computer Science** 2017 – 2022
Northwestern University Evanston, IL
 - Advisor: Prof. [Yan Chen](#)
 - Area: Networked System, Cloud Networking, Network Protocols
 - Thesis Topic: Enabling Robust and Secure Edge-to-Cloud Communications
 - GPA: 3.97/4.00
- **Master's in Computer Science** 2014 – 2017
Xi'an Jiaotong University Shaanxi, China
 - Advisor: Prof. [Chengchen Hu](#)
 - Area: Software Defined Networking (SDN)
 - Rank: 1st/89 GPA: 3.81/4.00 Average: 91.7
- **B.E. in Software Engineering** 2010 – 2014
Xi'an Jiaotong University Shaanxi, China
 - Rank: 1st/78 GPA: 3.94/4.00 Average: 92.4 (in Junior and Senior years)

Work Experiences

- **Alibaba Cloud** Shanghai, China
Senior Engineer, at Kangaroo Network Group Sep. 2022 – Present
 - Aiming to provide flexible, efficient networks: within containers, above VPCs, for agentic sandboxes
- **SRI International** Menlo Park, CA
Senior Research Intern, at Computer Science Laboratory Jun. 2021 – Sep. 2021
 - Advisor: Dr. [Vinod Yegneswaran](#) and Dr. [Phillip Porras](#)

- Real-time anomaly detection with P4 switches and reinforcement learning for NextG networks

- **SRI International**

Menlo Park, CA

Research Intern, at Computer Science Laboratory

Jun. 2020 – Sep. 2020

- Advisor: Dr. Vinod Yegneswaran
- Designed and implemented an enterprise-wide radio situational awareness system
- Passively collected radio traffics, including Wi-Fi and Bluetooth, analyzed by AutoML and nPrint

Research: Cloud Networks & SDN

- **Network Optimizing for Microservices/Serverless Cloud** 2020 – 2024

- Microservices/serverless architectures bring flexibility but introduce network communication delay
- Integrated **QUIC** into **OpenFaaS/K8s**, improved serverless network performance by 28%
- Designed a clean-slate serverless network architecture for direct function chain communications
- Designed systematic benchmarks for the microservices network performance measurement

Publication: *[FaaSTracker, TIFS 2025], [DirectFaaS, WWW'24], [QFaaS, SoCC'22]*

- **Generic Security Policy Enforcement System for SDN-based Cloud** 2017 – 2018

- Designed a **policy language** for resource protection and management of SDN-based Cloud
- Implemented in the **OpenDaylight** controller, and deployed on **OpenStack**

Publication: *[IWQoS'18] Lightweight Resource Protection and Management System for SDN-based Cloud*

- **Routing Policy for Solving Reactive Model Overhead of SDN** 2016 – 2017

- SDN control channel bandwidth is the major bottleneck of applying reactive forwarding
- Proposed a routing policy to reduce the SDN control channel bandwidth consumption up to 80%
- Implemented in the **Floodlight** controller under the **OpenFlow/P4** switch and **Open vSwitch**

Publication: *[ICNP'17] SoftRing: Taming the Reactive Model for Software Defined Networks*

Research: Formal Methods for Network Protocols

- **Formal Verification and Vulnerability Detection of LTE/5G Protocols** 2019 - 2020

- Used **TLA+** to formally specify the emergency call systems in **4G/5G** cellular network protocols
- Built a **complete cellular network testbed** (USRP, OpenAirInterface) for real-world verification
- Discovered serious availability and security issues in real-world, **acknowledged by major carriers**

Publication: *[MobiSys'21] Discovering Emergency Call Pitfalls for Cellular Networks with Formal Methods*

- **Formal Secure Configuration Search for Network Protocols** 2020 - 2022

- Traditionally, researchers use secure properties to verify a protocol is safe or find counterexamples
- We convert this decision problem into a search problem. Given the model and the properties, we aim to search the boundaries of the configuration space where the system is always secure and reliable
- We employ inductive generalization and improved IC3 algorithm to determine the space

Publication: *[Sigcomm'20 Poster] Network Protocol Safe Configuration Search in One Shot*

Publication List

- ***FaaSTracker: Efficient Cross-Layer Provenance Tracking of Serverless Applications with Multi-Source Correlation***
Qingyang Zeng, Lianjie Wu, **Kaiyu Hou**, Xue Leng, Yan Chen
IEEE Transactions on Information Forensics and Security (TIFS), 2025
- ***Property Guided Secure Configuration Space Search***
You Li*, **Kaiyu Hou***, Yunqi He, Yan Chen, Hai Zhou (*equal contribution)
International Conference on Information Security, 2024 (ISC'24)
- ***DirectFaaS: A Clean-Slate Network Architecture for Efficient Serverless Chain Communications***
Qingyang Zeng, **Kaiyu Hou**, Xue Leng, Yan Chen
ACM The Web Conference, 2024 (WWW'24)
- ***LANTERN: Layered Adaptive Network Telemetry Collection for Programmable Dataplanes***
Kaiyu Hou, Dhiraj Saharia, Vinod Yegneswaran, Phillip Porras
6th European P4 Workshop, 2023 (EuroP4'23)
- ***QFaaS: Accelerating and Securing Serverless Cloud Networks with QUIC***
Kaiyu Hou, Sen Lin, Yan Chen, Vinod Yegneswaran
ACM Symposium on Cloud Computing, 2022 (SoCC'22)
- ***Accelerate and Secure Serverless Networks with QUIC***
Kaiyu Hou, Sen Lin, Yan Chen, Vinod Yegneswaran
ACM 17th International Conference on Emerging Networking Experiments and Technologies, Poster, 2021 (CoNEXT'21)
- ***Discovering Emergency Call Pitfalls for Cellular Networks with Formal Methods***
Kaiyu Hou*, You Li*, Yinbo Yu, Yan Chen, Hai Zhou (*equal contribution)
ACM 19th International Conference on Mobile Systems, Applications, and Services, 2021 (MobiSys'21)
- ***Network Protocol Safe Configuration Search in One Shot***
You Li*, **Kaiyu Hou***, Hai Zhou, Yan Chen (*equal contribution)
ACM Special Interest Group on Data Communication, Poster, 2020 (SIGCOMM'20)
- ***State of the Art and Research Challenges in the Security Technologies of Network Function Virtualization***
Xiaochun Wu, **Kaiyu Hou**, Xue Leng, Xing Li, Yinbo Yu, Bo Wu, Yan Chen
IEEE Internet Computing, 2020
- ***CellScope: Automatically Specifying and Verifying Cellular Network Protocols***
Yinbo Yu, You Li, **Kaiyu Hou**, Yan Chen, Hai Zhou, Jianfeng Yang
ACM Special Interest Group on Data Communication, Poster, 2019 (SIGCOMM'19)
- ***A Lightweight Policy Enforcement System for Resource Protection and Management in the SDN-based Cloud***
Xue Leng, **Kaiyu Hou**, Yan Chen, Kai Bu, Libin Song, You Li
Computer Networks, Elsevier, 2019

- **COIN: A fast packet inspection method over compressed traffic**
Xiuwen Sun, Hao Li, Dan Zhao, Xingxing Lu, **Kaiyu Hou**, Chengchen Hu
Journal of Network and Computer Applications, Elsevier, 2019
- **SDNKeeper: Lightweight Resource Protection and Management System for SDN-based Cloud**
Xue Leng, **Kaiyu Hou**, Yan Chen, Kai Bu, Libin Song
IEEE/ACM 26th International Symposium on Quality of Service, 2018 (IWQoS'18)
- **SoftRing: Taming the Reactive Model for Software Defined Networks**
Chengchen Hu, **Kaiyu Hou** (1st student author), Hao Li, Ruilong Wang, Peng Zheng, Peng Zhang, Huanzhao Wang
IEEE 25th International Conference on Network Protocols, 2017 (ICNP'17)
- **Towards A Fast Packet Inspection over Compressed HTTP Traffic**
Xiuwen Sun, **Kaiyu Hou**, Hao Li, Chengchen Hu
IEEE/ACM 25th International Symposium on Quality of Service, 2017 (IWQoS'16)

Teaching Experiences

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|---|--------------|
| • CS 214: Data Structure and Data Management
<i>Teaching Assistant, Northwestern University</i> | Spring, 2022 |
| • CS 214: Data Structure and Data Management
<i>Teaching Assistant, Northwestern University</i> | Spring, 2021 |
| • CS 214: Data Structure and Data Management
<i>Teaching Assistant, Northwestern University</i> | Winter, 2021 |
| • CS 214: Data Structure and Data Management
<i>Teaching Assistant, Northwestern University</i> | Fall, 2020 |
| • CS 212, Mathematical Foundations of Computer Science
<i>Teaching Assistant, Northwestern University</i> | Spring, 2020 |
| • CS 340, Introduction to Networking (Best Teaching Assistant Award)
<i>Teaching Assistant, Northwestern University</i> | Winter, 2020 |
| • CS 214, Data Structure and Data Management
<i>Teaching Assistant, Northwestern University</i> | Fall, 2019 |
| • EECS 343, Operating Systems
<i>Teaching Assistant, Northwestern University</i> | Spring, 2019 |
| • EECS 340, Introduction to Networking
<i>Assistant Lecturer, Northwestern University</i> | Winter, 2019 |
| • EECS 340, Introduction to Networking
<i>Teaching Assistant, Northwestern University</i> | Fall, 2018 |
| • IT 458, Information Security and Assurance | Winter, 2018 |

Teaching Assistant, Northwestern University

- Computer Programming (C++) for CS Honored Class Fall, 2016
Teaching Assistant, Xi'an Jiaotong University

Professional Services

- Sub-reviewer of ACM ASIA-CCS (2021)
- Sub-reviewer of ACM CCS (2018, 2019)
- Sub-reviewer of IEEE ICDCS (2018)
- Reviewer of IEEE/ACM Transactions on Networking (ToN)

Awards

- **Bronze Medal**, ACM-ICPC Asia Regional Contest 2012, 2013, 2014
- **Silver Medal**, ACM-ICPC China Province Contest, Chengdu 2012, 2013
- **Meritorious Winner**, Mathematical Contest in Modeling (MCM) 2013

Honor

- **Terminal Year Cabell Fellowship** 2021
Northwestern University
- **Best Teaching Assistant Award** (*Peter and Adrienne Barris Outstanding TA*) 2020
Northwestern University
- **Outstanding Graduate (Cadre) Award** 2017
Xi'an Jiaotong University
- **Excellent Postgraduate Student Leader Award** (each year) 2015, 2016
Xi'an Jiaotong University
- **Outstanding Graduate Award** 2014
Xi'an Jiaotong University
- **Excellent Student Model Nomination** (16/13000) 2013
Xi'an Jiaotong University
- **Google Excellence Scholarship** 2013
Awarded to 3 students from each of 20 top Chinese universities
- **Excellent Student Award** (each year) 2011, 2012, 2013
Xi'an Jiaotong University

Social Activities

- **Student President** of the Computer Science Dept., (Class 2017) 2014 – 2017

Xi'an Jiaotong University

- **Student Councilor**, the Student Congress of ChungYing College
Xi'an Jiaotong University 2013 - 2014
- **Chair**, the ACM-ICPC Club
Xi'an Jiaotong University 2012 - 2013
- **Science and Education Deputy Director** of the Student Union
Xi'an Jiaotong University 2011 - 2012
- **Co-Founder & the 2nd Store Manager** of *Bingo Cafe*
ChungYing College, Xi'an Jiaotong University 2010 - 2013

Skills

- QUIC Protocol, TCP/IP Protocols, 3GPP Cellular Network Protocols (NAS layer), 802.11 Protocols
- Serverless Computing, Serverless Network Architecture, Cloud Networks, SDN, P4
- Formal Methods for Network Protocols, TLA+, IC3, NuSMV
- Python, Go-lang | Container, Microservices, Kubernetes, CNI